HSR-PE-07 SOP: Resolving QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on Cisco PE Router

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_7868

Description

This Standard Operating Procedure (SOP) addresses a Major alarm ALARM_7868 on the PE Router HSR-PE-07 located at HSR Layout Access, which is a Cisco router operating in a PE role. The failure domain in question pertains to QoS / Congestion Event, specifically microbursts, interface tail drops, and QoS queue starvation. This scenario arises due to the complex interplay between the Cisco PE's Quality of Service (QoS) mechanisms, which prioritize traffic based on various parameters such as class of service (CoS), differentiated services code point (DSCP), and IP precedence, and the congestion control mechanisms that regulate the traffic flow on the interface. When the PE router encounters a situation where the incoming traffic exceeds its processing capacity, it resorts to tail drop or random early detection (RED) algorithms to mitigate congestion. However, in certain cases, this may lead to microbursts and subsequent QoS queue starvation, triggering the ALARM_7868. To resolve this issue, it is essential to identify the root cause, which could be attributed to a variety of factors including misconfigured QoS policies, inadequate interface capacity, or malfunctioning hardware components.

Pre-requisite Checks

- check: Verify that the redundant link is operational and functioning correctly.

- check: Confirm that the remote hands have been dispatched to HSR Layout Access to facilitate necessary actions.

- check: Ensure that the Cisco PE router HSR-PE-07 is accessible via the console or SSH for command-line interface (CLI) access.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate interface capacity	Medium
Malfunctioning hardware components	Low

Diagnostic Commands & Expected Outputs

```
Command: show queueing queue , Expected Output: Displays the queueing information for the specified interface, including the number of packets in the queue, the maximum size of the queue, and the current queueing mode., Description: This command isolates the queueing behavior on the specified interface, helping to identify potential
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issues related to microbursts or QoS queue starvation.

Command: show policy-map interface , Expected Output: Displays the policy map configuration applied to the specified interface, including the class maps, match criteria, and action statements., Description: This command diagnoses the QoS policy configuration applied to the interface, enabling the identification of potential misconfigurations or mismatches between the policy map and the incoming traffic.

Command: show interface | include drops, Expected Output: Displays the interface statistics, including the number of packets dropped due to congestion or other reasons., Description: This command isolates the interface statistics related to packet drops, helping to identify potential issues with interface capacity or congestion control mechanisms.
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Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: configure terminal delete policy-map delete class-map , Description: This step involves reviewing and correcting the QoS policy configuration to ensure that it aligns with the intended traffic prioritization and shaping requirements. This may involve deleting or modifying existing class maps and policy maps, or creating new ones as necessary.

Step 2: Step: Adjust interface capacity or congestion control mechanisms, Command: interface bandwidth queue-limit , Description: This step involves adjusting the interface capacity or congestion control mechanisms to mitigate potential issues related to microbursts or QoS queue starvation. This may involve configuring a new bandwidth value or queue limit on the interface.

Post-Check Verification

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Command: show queueing queue , Expected Output: Displays the queueing information for the specified interface, including the number of packets in the queue, the maximum size of the queue, and the current queueing mode., Description: This command verifies that the queueing behavior on the specified interface has been corrected, and that the number of packets in the queue is within expected limits.

Command: show policy-map interface , Expected Output: Displays the policy map configuration applied to the specified interface, including the class maps, match criteria, and action statements., Description: This command verifies that the QoS policy configuration on the interface has been corrected, and that the policy map configuration aligns with the intended traffic prioritization and shaping requirements.
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Roll-back Procedures

- Step: Restore the original QoS policy configuration, Command: configure terminal copy running-config startup-config reload, Description: This step involves restoring the original QoS policy configuration from the startup configuration file, and reloading the router to ensure that the changes take effect.

- Step: Revert changes to interface capacity or congestion control mechanisms, Command: interface bandwidth queue-limit , Description: This step involves reverting the changes made to the interface capacity or congestion control mechanisms, and restoring the original values.